

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

Department of Electrical Engineering - ICDT

PROJECT REPORT

Development and Testing of a Charger with a Custom-Designed Enclosure

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1 Introduction

This project is about building a mobile phone charger using basic electronic components. The 230 V AC supply is stepped down to 12 V AC using a transformer, converted to DC using four PN diodes, filtered using a capacitor, and regulated using a voltage regulator. The final output is given through a USB-A port for charging a phone. The circuit is assembled on a perfboard.A 3D-printed case is also designed and made to safely house the circuit and give the charger a neat and compact finish.

2 Project Objectives

- To build a Functional mobile phone charger on a perfboard.
- To convert 230 V AC supply into a stable DC output.
- To understand the working of the transformer, rectifier, capacitor, and regulator, and the process of AC to DC conversion.
- To assemble the circuit and solder neatly on a perfboard.
- To design and 3D print a case for the charger.
- To test the charger and verify its output voltage.
- To gain hands-on experience in circuit assembly, with electronics, soldering, and 3D printing.

3 Component Overview

The charger is implemented using discrete hardware components distributed for the laboratory project. This section details the list of materials used in the physical assembly.

3.1 List of Materials

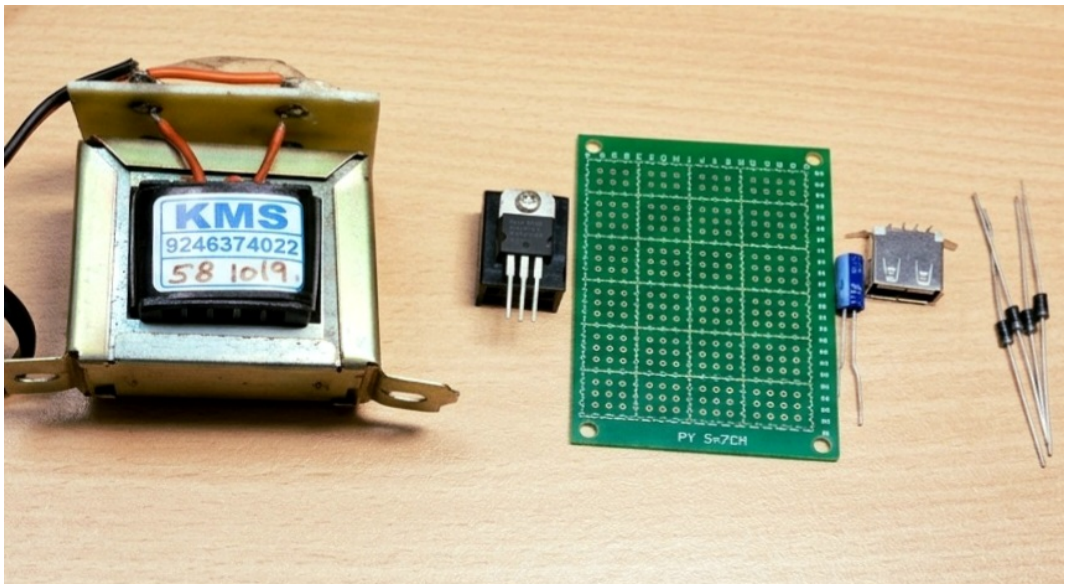


Figure 1: Components for Charger

Table 1 outlines the specific parts received and integrated into the perfboard setup.

Table 1: List of Components

S.No.	Component Name	Quantity
1	Step-Down Transformer	1
2	Voltage Regulator IC	1
3	Perf Board	1
4	Capacitor(100 μF/25V/85°C)	1
5	USB Type-A Port	1
6	PN Junction Diodes	4
7	Connecting Wires and Solder	as needed

3.2 Brief Overview of Components

- **`Step-down Transformer:`** Reduces the 230 V AC mains voltage to a lower 12 V AC voltage suitable for the charger circuit.
- **`Voltage Regulator:`** Maintains a stable DC output voltage suitable for charging a mobile phone.There is also a Heat sink to it, which helps remove excess heat produced by the voltage regulator during operation.
- **`Perf-Board:`** Used to mount and permanently connect the electronic components through soldering.
- **`Capacitor:`** A capacitor stores and releases electrical charge and is commonly used to smooth voltage, filter noise, and reduce fluctuations in electronic circuits.
- **`PN Diodes:`** It allows current to flow mainly in one direction and is commonly used for rectification, switching, and protection in electronic circuits.

4 Working Principle

- The 230 V AC mains voltage is first supplied to the step-down transformer. It reduces the high input voltage to approximately 12 V AC, making it suitable for the low-voltage charging circuit.

- **`Bridge Rectifier:`** A full-wave bridge rectifier is a circuit made using four PN junction diodes connected in a bridge arrangement.

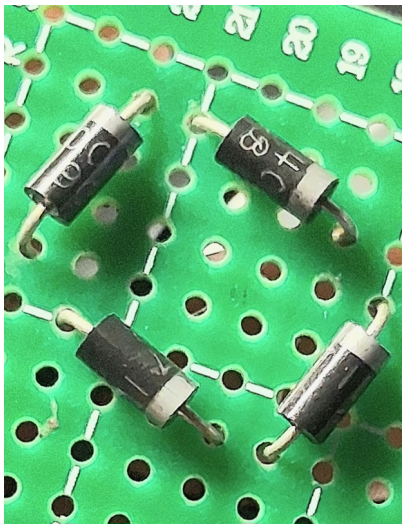


Figure 2: Full Wave Bridge Rectifier

It is used to convert AC voltage into pulsating DC voltage. In our charger, it receives the 12 V AC from the transformer and converts it into pulsating DC. Two diodes conduct during each half-cycle, keeping the current flowing in the same direction through the circuit. The resulting pulsating DC is then passed through the capacitor, which smooths the voltage before it reaches the regulator.

- The pulsating DC is passed through a capacitor. The capacitor charges and discharges continuously, reducing the fluctuations in the rectified voltage and producing a smoother DC output.

- The filtered DC voltage is then given to the voltage regulator. It maintains a stable output voltage required for the phone charger. The heat sink helps dissipate the heat produced by the regulator during operation.

- The regulated DC output is connected to the USB-A connector. A suitable charging cable can then be connected to the USB port to supply power to the mobile phone.

In summary, we can say the working principle as:

230 V AC → Step-down Transformer → 12 V (RMS) AC → Bridge Rectifier → 17 V(peak) DC → Capacitor → Voltage Regulator → 5 V DC → USB-A → Phone

4.1 Circuit Diagram

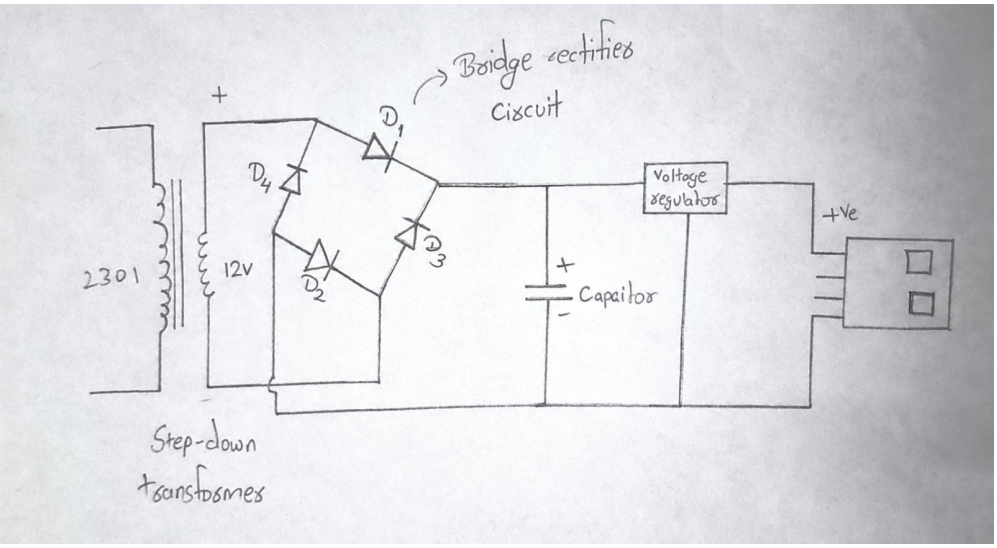


Figure 3: Circuit Diagram of Charger

5 Circuit Assembly on Perf-Board

5.1 Component Placement

The components were placed on the perfboard, according to the circuit diagram,keeping the connections as compact and organized as possible and the assembled perfboard looks like the Figure.5 .

5.2 Soldering the Components

The components were soldered onto the perfboard according to the circuit connections.

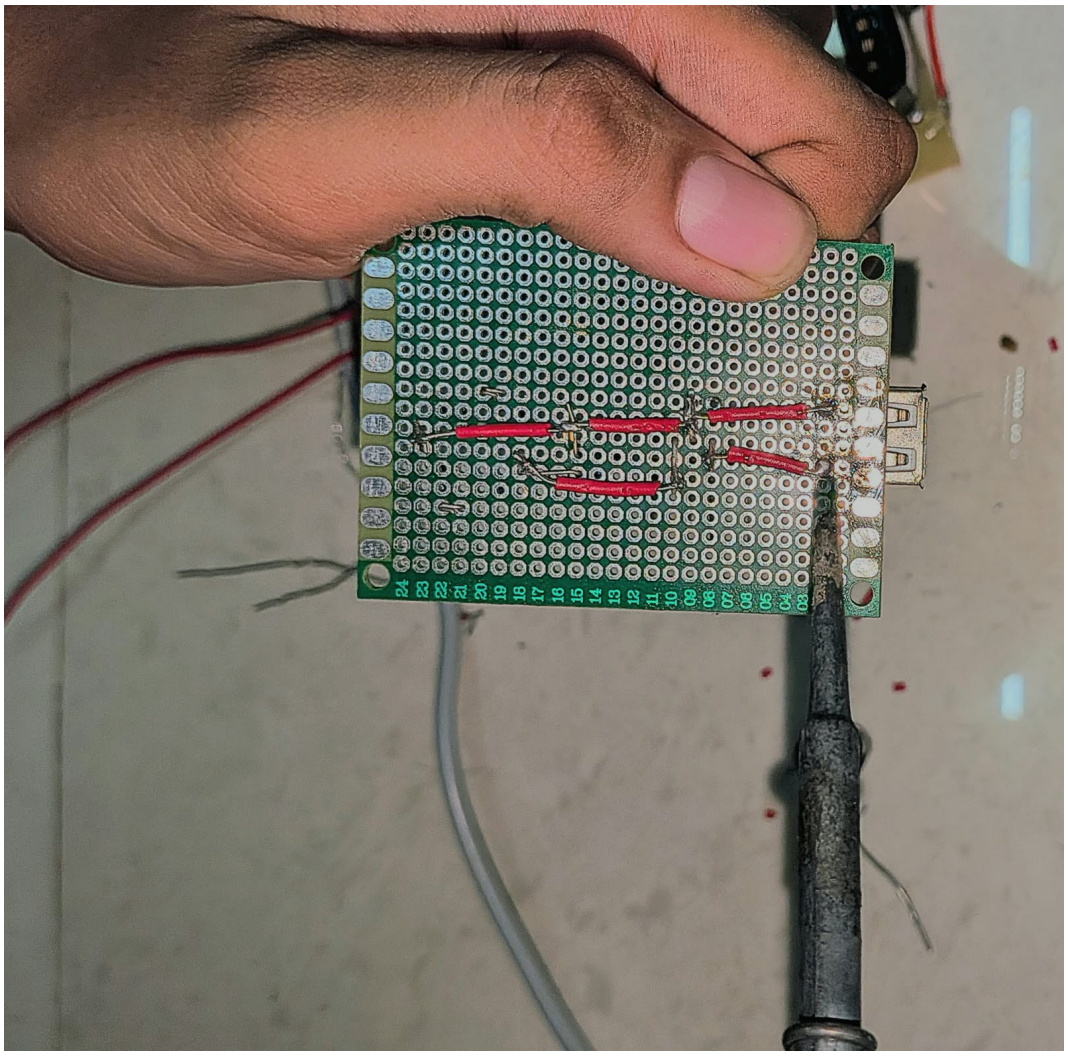


Figure 4: Soldering the Components

During soldering, care was taken to make the joints secure while avoiding excessive solder and unnecessary wire ends. The connections were kept clean and organized to give the circuit a neat overall appearance.

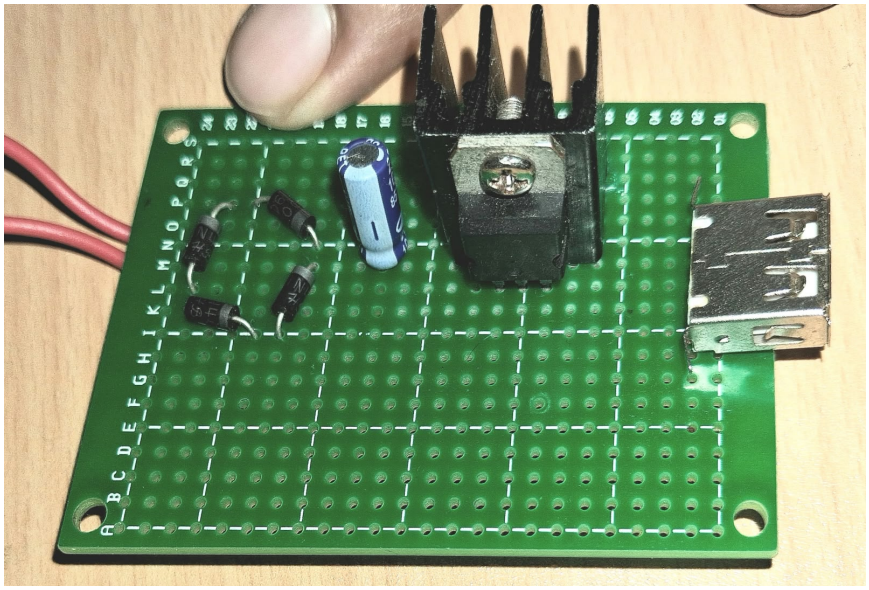


Figure 5: Soldered Top

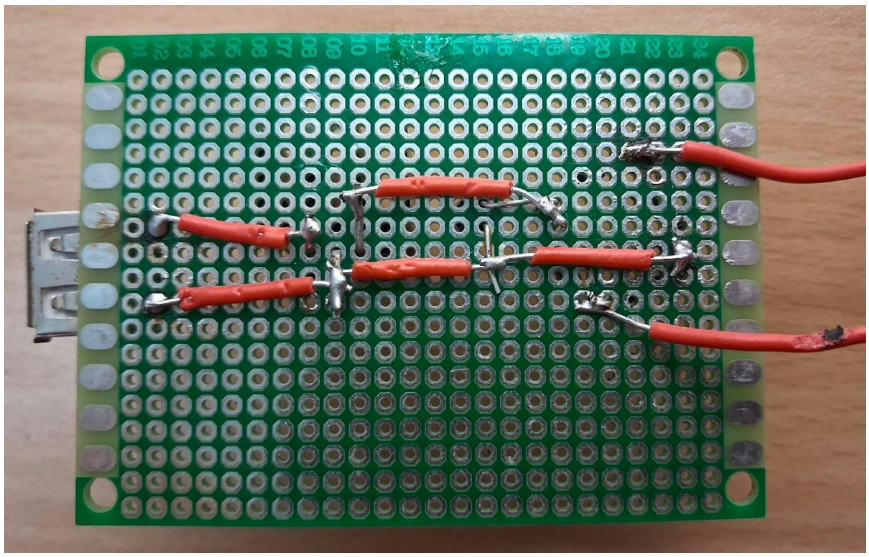


Figure 6: Soldered Bottom

6 Testing and Voltage Measurements

After soldering the components, we measured the voltages at different points of the circuit using a multimeter to make sure everything was working properly.

We checked the voltage at different stages of the circuit and also measured the final output from the USB-A port.

Transformer Output

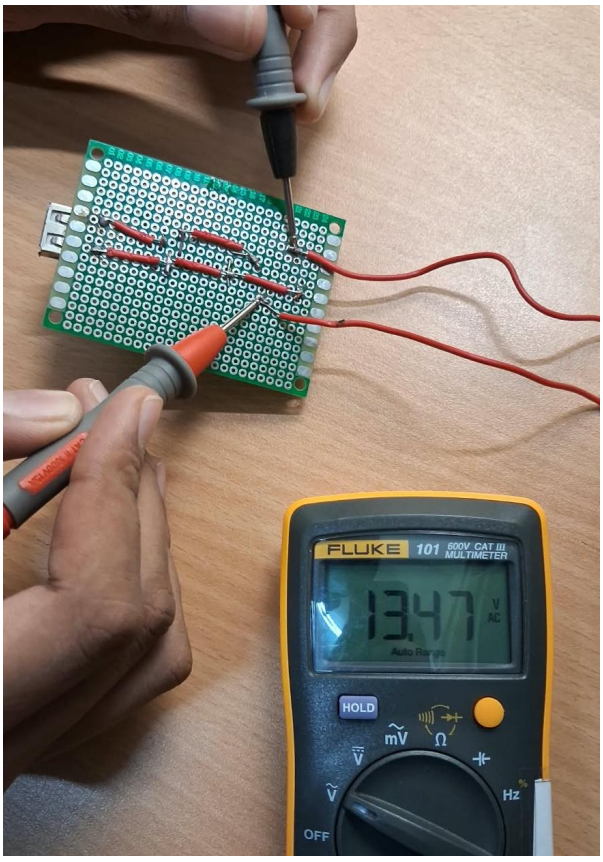


Figure 7: Transformer Output

Rectifier Output

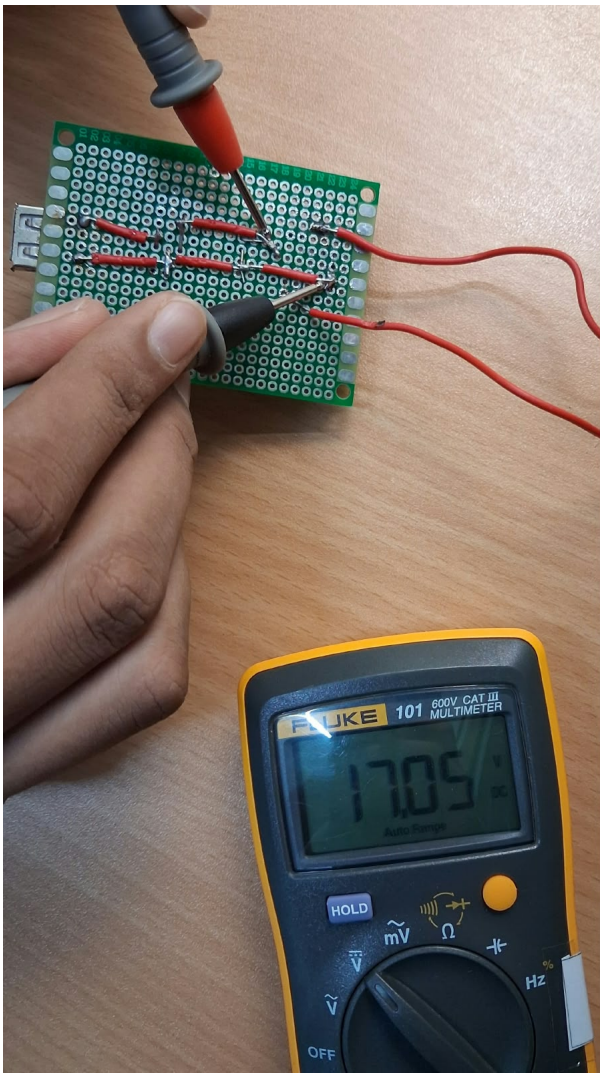


Figure 8: Rectifier Output

Voltage Regulator Output

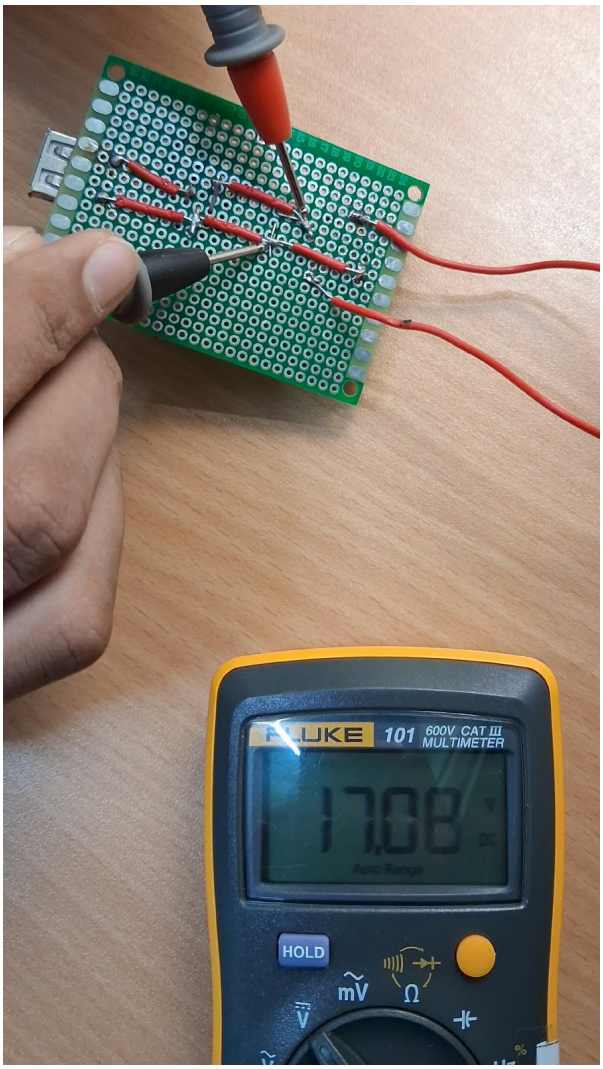


Figure 9: Regulator Output

USB Output

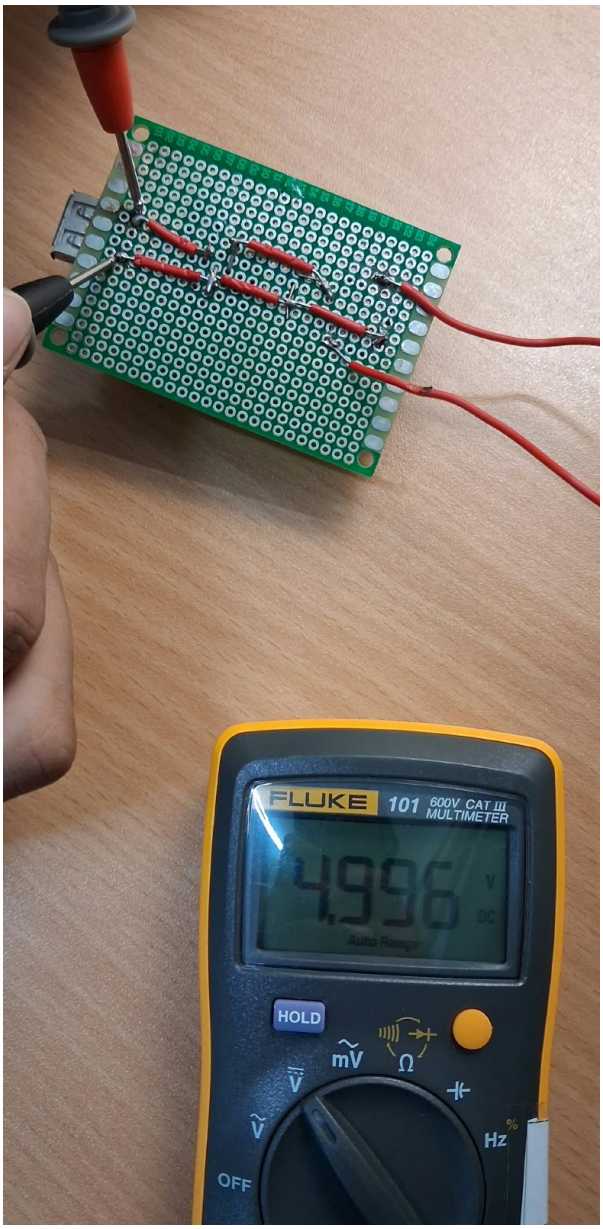


Figure 10: USB Output

These readings helped us confirm that the circuit was working, safely, as expected.

7 Case Designing & 3D Printing

This was easily the most frustrating part of the project. Getting the measurements, component clearances, and overall shape right took several attempts. Turns out, making the charger case is harder than the Charger itself!

7.1 Initial Rough Design

Before starting the 3D modelling, we first discussed the basic structure of the case and made a rough sketch on paper. The sketch helped us visualize the approximate shape, component placement, and openings required for the charger. The final design changed considerably from this initial idea as we explored different designs and considered the actual dimensions of the circuit.

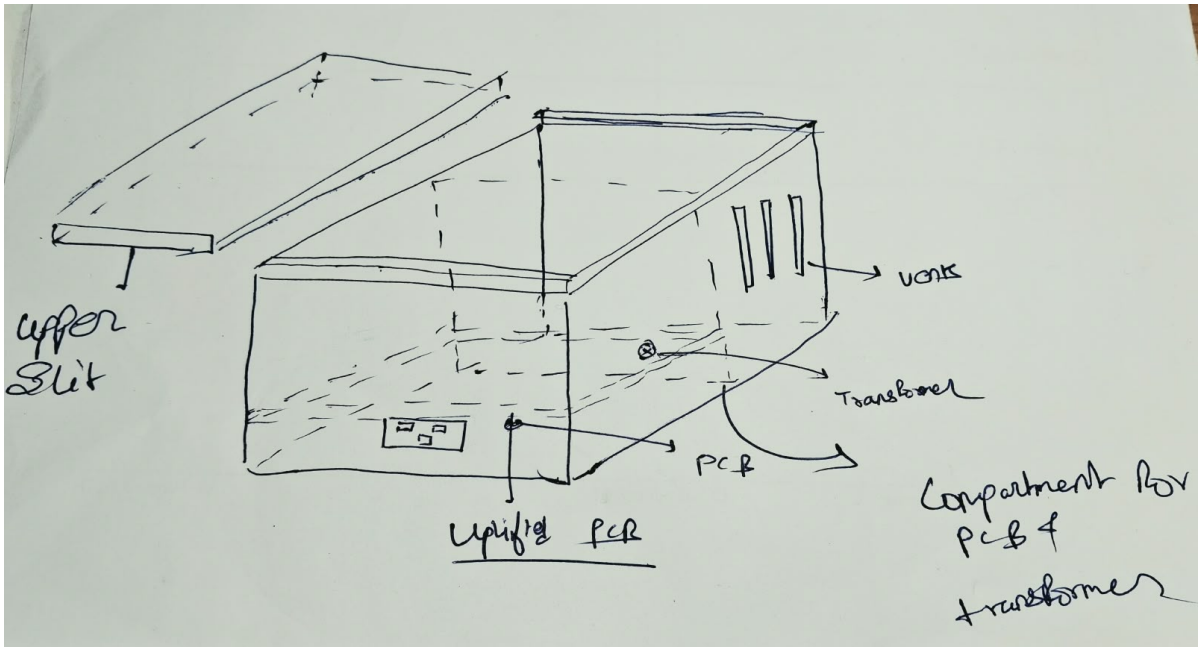


Figure 11: rough sketch of design

7.2 Choosing the Modeling Software

Before beginning the actual modeling of the charger case, we explored different 3D design software to understand how they worked and mainly to check which was more beginner friendly. We experimented with basic tools, shapes, and modelling operations in Blender, FreeCAD, and OpenSCAD.

- **Blender:** We experimented with it to understand its basic tools and 3D modelling workflow.We also tried to make the outline for the case in it, as shown below.

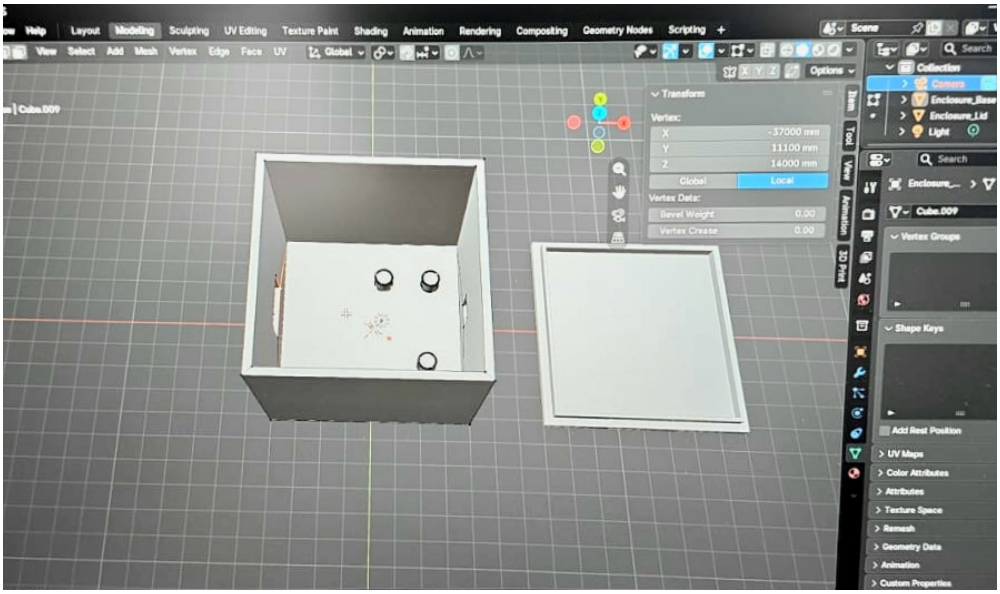


Figure 12: Blender Testing

- **FreeCAD:** We spent some time experimenting with it and tried familiarising ourselves with it's basic 3D modeling tools. But the UI felt less beginner-friendly / complex, as compared to Blender and OpenSCAD.

- **OpenSCAD:** It felt easier to work with, and we found it more convenient to make and modify the case design and use AI to vibe code for further modifications.

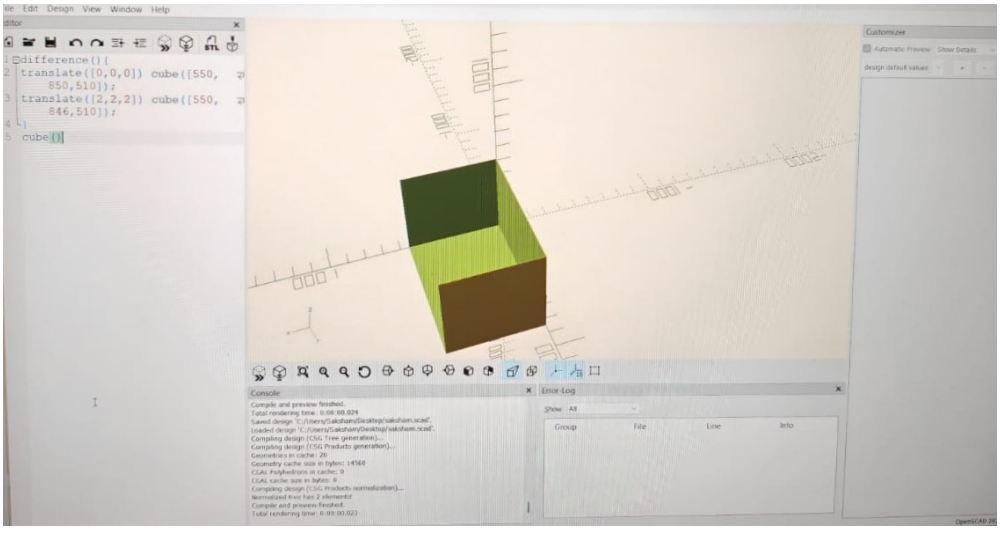


Figure 13: OpenSCAD Testing

7.3 Final Case Design

The following image shows our final case design, made using OpenSCAD.

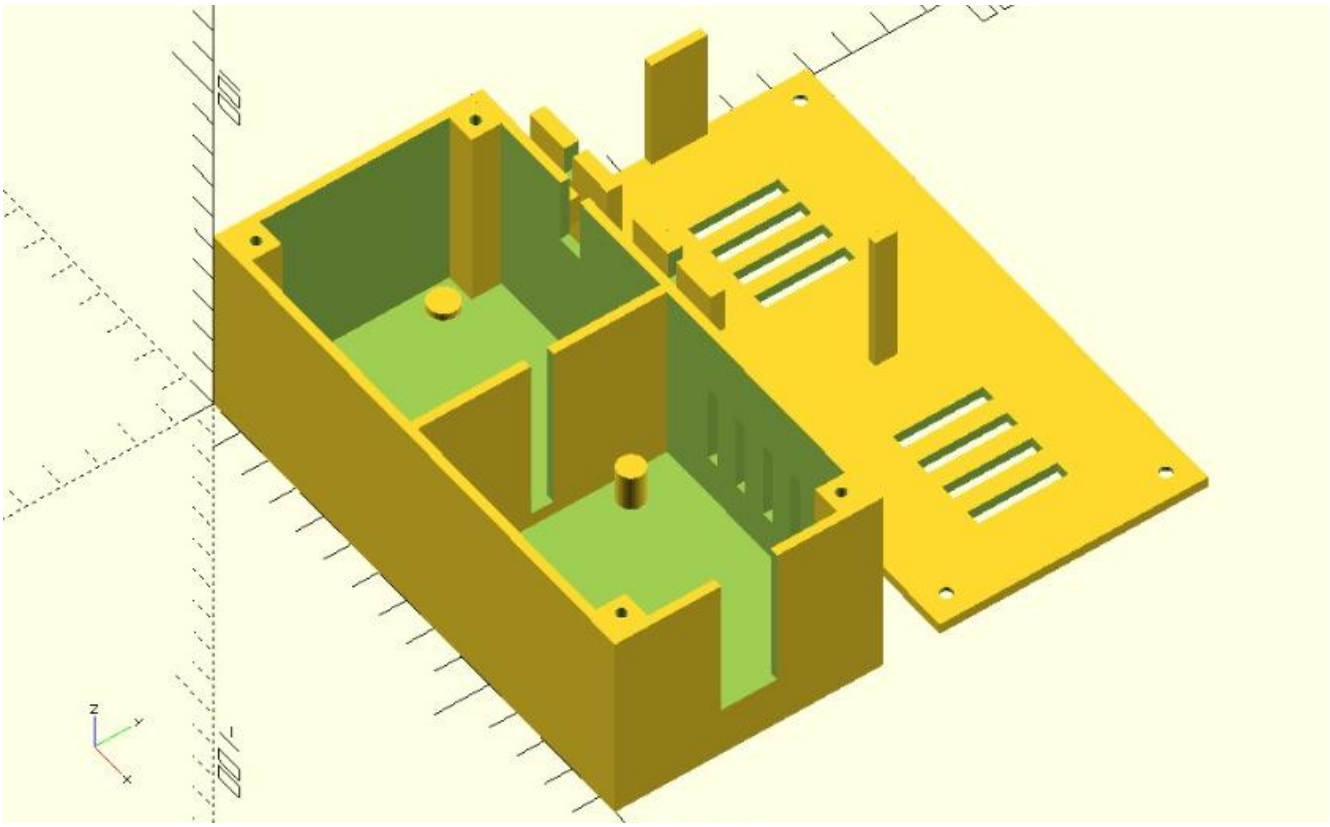


Figure 14: Final Case Design

The initial rough sketch was mainly for the outline and as a starting point. The design changed considerably as we experimented with the model and considered the practical requirements of the charger. Different shapes, dimensions, and arrangements were tried before arriving at the final design.

Particular attention was given to the dimensions of the enclosure so that the charger components could fit inside comfortably. The dimensions were adjusted several times as the model developed.We also didn't forget to account for the component clearance space.

We understood how crucial taking measurements of the components is for 3D printing.

7.4 Major Design Modifications

- **Ventilation strips**

  Since some power is dissipated as heat by the components while using the charger, a heat sink is provided to help transfer this heat away from the components. Ventilation strips were added to the enclosure to allow this heat to dissipate and maintain adequate airflow around the heat sink.

- **Case Fastening Mechanism**

  The initial design incorporated a sliding mechanism for opening and closing the case.

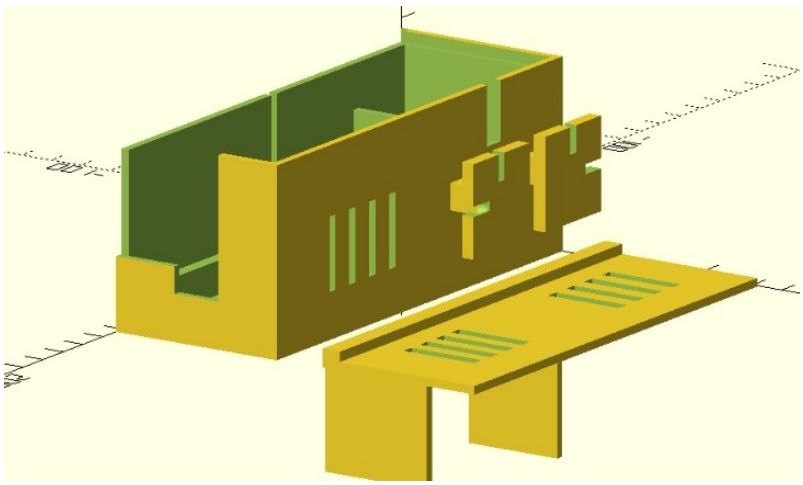


Figure 15: Sliding Prototype

  However, based on feedback from our seniors regarding the practical difficulties and reliability of implementing the sliding mechanism, this feature was removed in the revised design. A screw-based fastening system was adopted instead to provide a simpler and more secure mechanical connection between the case components.

- **Cord Management Pegs**

  Two pegs were added to the side of the case to allow the long cord of the step-down transformer to be wrapped around them, keeping the cord neatly organized and secured. We also tried to make a hook type thing for the plug to rest too.

- **PCB-Transformer Separator**

An internal separator wall was added between the PCB and transformer to keep the two components physically separated. It creates separate compartments and keeps the internal layout organized. It also helps prevent heat and other effects from one component from directly affecting the other compartment.

8 Problems Faced

Compared to the clock project, this project was relatively simpler and involved fewer difficulties. However, we still faced a few challenges during the circuit assembly, testing, and enclosure development. These challenges helped us identify areas for improvement and make the necessary changes to the final design.

8.1 Soldering and Circuit Assembly

- Soldering was initially a challenge because most of us had little experience with it. One of our teammates has some prior experience with soldering, so he(Ganesh) handled most of the soldering work and helped us learn the process.
- We came to know that some joints were weak when they broke during handling and resoldered them properly to make the connections stronger and more reliable.
- One of the PN junctions was soldered in the wrong direction. It took us some time to find the mistake, after which we desoldered it and soldered it again correctly.
- The step-down transformer we initially received made a loud buzzing sound during operation, which raised concern about whether it was working properly. We decided not to continue using it and replaced it with another transformer, which worked properly.

8.2 3D Design

- The main difficulty we faced during the 3D designing part was deciding on the final design of the enclosure. We discussed several ideas and considered different ways of arranging the components, fastening the case, managing the transformer cable, and providing ventilation. The initial design changed considerably as we discussed the practical requirements of the charger. Although the actual modelling process became easier with the help of AI tools such as Gemini, deciding on a suitable and practical design took considerable discussion and several changes.

9 Learnings and Experience

- **Soldering**
  Before this project, we thought soldering was a very difficult and dangerous process, almost like welding. But once we actually started doing it, we realised that it was much simpler than we had imagined. By the end of the project, we got a basic idea of how to solder components properly and make good connections.

- **Circuit Understanding**
  During this project, we also learned how to use a multimeter for the first time. We used it to check the terminals of different components and make sure the connections were working properly. This also helped us understand the circuit better and learn how to check whether everything was working correctly.

- **3D Designing**
  Before starting the 3D designing part, we were a little unsure about it. From the YouTube videos we had seen, we thought we would need a vernier caliper and heavy machinery to make an enclosure. We also thought getting the right measurements would be difficult. But after trying the software ourselves, we realised it was much simpler than we expected. We tried Blender, FreeCAD and OpenSCAD before choosing OpenSCAD for the final design.

  We also learned that designing an enclosure is not just about making it look good. We had to measure the components so they would fit properly and consider their placement, ventilation, access and the transformer cord.

10 Completion and Conclusion

10.1 Charger Functional

After completing the circuit, we tested the final charger first by multimeter and after reassurance, by connecting it to a phone. The charger was able to charge the phone successfully, showing that the final circuit was working properly, as shown below.

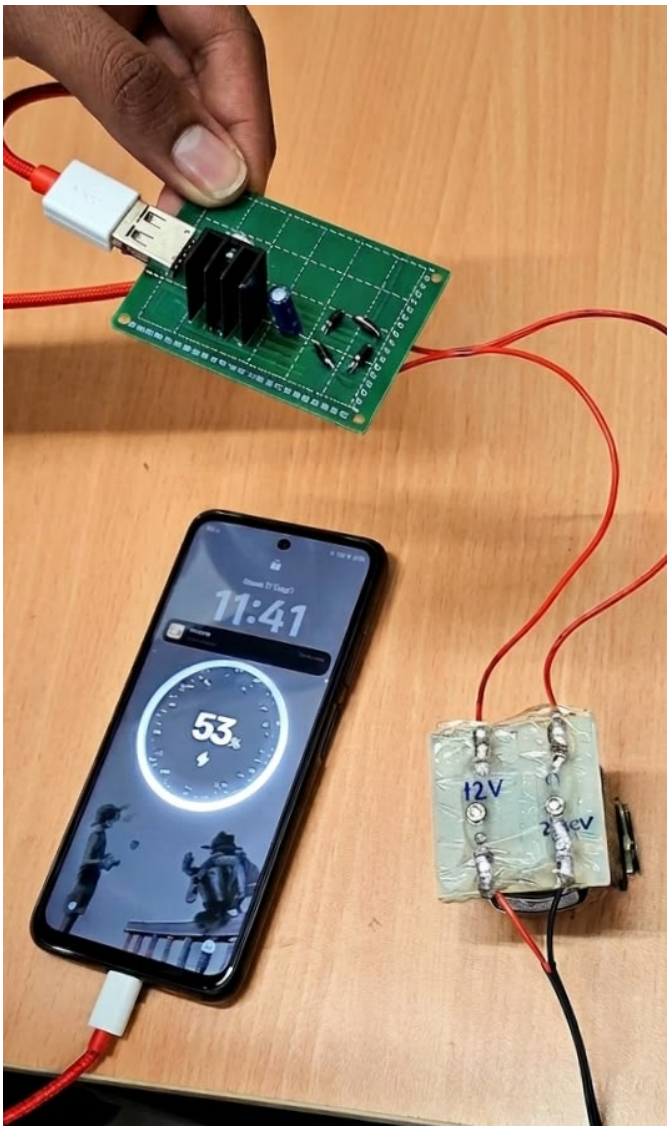


Figure 16: Working Charger

10.2 Conclusion

Overall, we were able to successfully complete our charger and its enclosure. We started with a basic idea and slowly worked on it until we had a working product. The design also changed along the way as we understood what was actually needed for the charger and its components.

The best part of the project was seeing the final charger actually work and being able to use it to charge a phone. It was a good experience for us because we got to work on both the circuit and the physical design of the product. Overall, the project helped us understand how an idea can be turned into a working product through practical work and trying things out ourselves.